



# SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category - I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

**Course Name:** Statistics and Probability  
**Course Code:** TE7689  
**Faculty:** Engineering  
**Course Credit:** 3  
**Course Level:** 1  
**Sub-Committee (Specialization):** Applied Sciences  
**Learning Objectives:**

The students will be able to:

1. Understand the statistical data and their graphical representation wherever relevant.
2. Interpret the data using appropriate measures of central tendency.
3. Find measures of dispersion and compare two series using coefficients of variation.
4. Find coefficient of correlation and rank correlation.
5. Explain the line of best fit and estimate the value of dependent variable using regression analysis.
6. Define the concept of random variables and find the probability distribution.

**Books Recommended:**

| Book                                                                    | Author                                                             | Publisher                                              |
|-------------------------------------------------------------------------|--------------------------------------------------------------------|--------------------------------------------------------|
| Fundamentals of Statistics                                              | A. M. Gun, M.K. Gupta and B. Das Gupta                             | World Press Private Ltd (1 August 2013)                |
| Probability and Statistics                                              | Murray Spiegel, John Schiller, R. Alu Srinivasan, Debasree Goswami | McGraw Hill Education; 3rd edition (1 July 2017)       |
| Schaum's Outline of Probability, Random Variables, and Random Processes | Hwei Hsu                                                           | McGraw-Hill Education; 4th edition (30 September 2019) |
| Statistics                                                              | S. P. Gupta                                                        | Sultan Chand and Sons Since 1950 (1 January 2021)      |
| Statistics With R Programming                                           | T. Murali Mohan                                                    | Sultan Chand Publications, 2020.                       |

**Course Outline:**

| Sr. No. | Topic                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Actual Teaching Hours | Contact Hours Equivalence |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|---------------------------|
| 1       | <b>Statistical Techniques:</b><br>Definition of statistics, Importance, scope and limitations of statistics, Primary and secondary data, Attributes and variables, Discrete and continuous variables, Graphical representations of Histogram, Ogive curves, stem and leaf charts. Definition of grouped and ungrouped data, Arithmetic mean, median and mode, Properties and relations of mean, median and mode, Range, Quartile, Quartile deviation, Percentile, Mean deviation, Standard deviation, Variance, Coefficient of variation, Moments, Skewness and kurtosis, Requisites of good measures of dispersion, Absolute and relative measures of dispersion, Merits and demerits of statistical measures. | 12                    | 12                        |
| 2       | <b>Bi-variate analysis:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 9                     | 9                         |

|              |                                                                                                                                                                                                                                                                                                                       |           |           |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------|
|              | Correlation: Introduction of correlation, Karl Pearson's coefficient of correlation, short-cut method, Step deviation method for coefficient of correlation, Rank correlation. Regression: Introduction of regression, Lines of regression, Properties of regression coefficient, Angle between two regression lines. |           |           |
| 3            | <b>Probability distributions:</b><br>Introduction to probability, Random variables, Discrete probability distribution, Binomial and Poisson distributions, Continuous probability distribution, Normal distribution, Uniform distribution, Exponential distribution, Joint probability distribution.                  | 9         | 9         |
| 4            | <b>Tutorial:</b><br>Tutorial will be conducted on all topics                                                                                                                                                                                                                                                          | 15        | 15        |
| <b>Total</b> |                                                                                                                                                                                                                                                                                                                       | <b>45</b> | <b>45</b> |

**Pre Requisites:**

None

**Evaluation:**

A. Continuous Assessment

1. Unit Tests
2. Tutorials
3. Class Tests
4. Assignments

B. End Semester Examination

1. Written Examination

**Pedagogy:**

1. Interactive sessions

2. Class room teaching

3. Tutorials

4. Assignment

**Expert:**

Dr. Arundhati Warke, Professor, Deputy Director Academics, SIT

Dr. Ramesh Katta, Assistant Professor, SIT